CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B.Tech Examination (IT)

May 2012

IT302: Design & Analysis of Algorithm

Date: 02.05.2012,Wendesday **Time: 10:00 a.m to 1:00 p.m** **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(a) How to choose an algorithm for solving a single problem, when you have multiple algorithms for solving same problem are available? **2**

(b) Arrange following functions in n in increasing order: **2**

$2^n, \log_2 n, n^3, n^2 \log_2 n, 3^n$

(c) Write an algorithm to find the factorial of a given number. What is the running time of this algorithm? **3**

Q-2

(a) Give similarities and differences between backtracking and branch-and-bound **4**

(b) Explain Greedy-choice property and Optimal substructure property by an example. **3**

(c) Show that the running time of quick sort is O (n log n) when the array A is not sorted. **5**

(d) Write and explain general recurrence equation used to analyze the behaviour of divide and conquer algorithm. **2**

OR

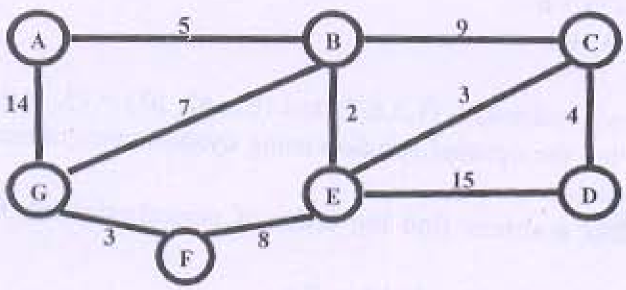
Q-2

(a) Write the recurrence equation for merge sort using divide and conquer approach. Also write its best case and worst case time complexity using recurrence. **4**

(b) Check the following equalities. **4**

1. $n^2 = O(n^3)$
2. $2^n = \Omega(2^{n+1})$

(c) Which edges form the minimum spanning tree (MST) of the below graph using kruskal algorithm? **6**



Q-3 **Attempt Any Two.** **14**

(a) Define the *"n queen problem"*. Write an algorithm using backtracking to solve *"n queen problem"* and using the algorithm find the solution for 5 queen problem.

(b) We have set of n jobs to execute, each of which takes unit time. At any time T=1, 2, we can execute one job. Job i earn us a profit Pi > 0 if and only if it is executed no late than time di (deadline). Develop a greedy algorithm to solve the above problem. Run the algorithm for n=4 and the following values.

i	1	2	3	4
p_i	50	10	15	30
d_i	2	1	1	7

(c) Sort the given numbers using quick sort: 25,24,13,45,17,31,96,50. Also determine asymptotic cost of Divide, Conquer and Combine.Choose first element as a pivot element.

SECTION-II

Q-4

(a) Explain O and Ω asymptotic notations used in analysis of algorithm in brief. 2

(b) Show the comparisons the naïve string matcher makes for the pattern P=0010 in the text T=000010001010001. 2

(c) Is dynamic programming a Top-Down or a Bottom-Up technique? Why? Explain with an example. 3

Q-5

(a) Solve the following task assignment problem instance for minimization. 6

	1	2	3	4
a	4	7	3	2
b	2	6	1	5
c	3	9	4	1
D	6	8	1	7

(b) Define the given terms : P Class,Np Class,Np Complete Class,NP Hard Class 4

(c) Solve the following recurrence exactly. 4

$t_n = n$ if n=0 and 1

$t_{n-1} + t_{n-2}$ otherwise

OR

Q-5

(a) Explain Rabin Karp Algorithm for string matching with suitable example. 5

(b) Define Time Complexity. Explain with the help of a suitable example Best case, Worst Case and Average Case Time Complexity of an algorithm 5

(c) State Master Theorem and Solve following recurrence using Master Theorem. 4

1. $T(n) = 2T(n/2) + n$
2. $T(n) = T(2n/3) + 1$

Q-6 Attempt Any Two. 14

(a) Consider n=3, (w1, w2, w3,w4) = (1,2,6,7) and (p1, p2, p3) = (3, 5, 8,11) and Capacity of Knapsack is W=9. Find the optimal solution using dynamic programming for 0/1 knapsack problem using given data.

(b) For the following four matrices find the order of parenthesization for the optimal chain multiplication.

A1 = 10 x 5, A2 = 5 x 3, A3 = 3 x 2, A4 = 2 x

(c) Determine the Logest Common Subsequence of the given two string : S1 : *ALGORITHM* S2: *LOGERITHM*. Show all the steps.